MATH 345 Skeleton Notes
Unit 4: Orthogonality and least squares
Section 4.1: Orthogonality and orthogonal bases

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Recall from Vectors that the dot product is symmetric and distributes over linear combinations, by Properties of the dot product. The norm satisfies $\|\mathbf{x}\|^2 = \mathbf{x} \cdot \mathbf{x}$, as noted in the referenced item. Two vectors are orthogonal when their dot product is zero. A nonzero vector can be normalized by dividing it by its norm.

Note: Dot-product notation and inner products

We have used $\mathbf{u} \cdot \mathbf{v}$ and $\mathbf{u}^T\mathbf{v}$ to denote the dot product. We will also use the notation $\langle \mathbf{u}, \mathbf{v} \rangle$. For real vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ written as column vectors, these notations mean the same thing:

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T\mathbf{v} = \langle \mathbf{u}, \mathbf{v} \rangle.$$

The dot product is also sometimes called an inner product. In Abstract vector spaces and polynomial approximation, we will generalize inner products to other vector spaces, including spaces of polynomials.

Orthogonal sets

We now study collections of mutually orthogonal vectors and how they provide convenient bases and coordinate formulas.

Definition: Orthogonal and orthonormal sets

A set of vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ is called an orthogonal set if

the set does not contain the zero vector, and $\mathbf{x}_i \cdot \mathbf{x}_j = 0$ for $i \neq j$. A set $\{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$ is called orthonormal if it is orthogonal and if in addition, $\|\mathbf{x}_i\| = 1$ for each i.

Note that $\{\mathbf{f}\}$ is an orthogonal set if $\mathbf{f} \neq \mathbf{0}$. The standard basis (see Standard basis) is an important orthonormal set in $\mathbb{R}^n$; here we may take $\mathbf{q}_i = \mathbf{e}_i$ for $i = 1, \ldots, n$.

Example: Checking orthogonality and normalizing vectors (U4-LO1)

Consider the following vectors in $\mathbb{R}^4$, written as column vectors:

$$\mathbf{f}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix} \quad \mathbf{f}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 2 \end{bmatrix} \quad \mathbf{f}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad \mathbf{f}_4 = \begin{bmatrix} -1 \\ 3 \\ -1 \\ 1 \end{bmatrix}.$$

Verify that these vectors form an orthogonal set in $\mathbb{R}^4$. Then find the orthonormal set $\{\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3, \mathbf{q}_4\}$ obtained by normalizing these vectors, i.e., multiplying each vector by an appropriate constant to obtain a unit vector.

To verify that $\{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3, \mathbf{f}_4\}$ form an orthogonal set, we need to check that $\mathbf{f}_i \cdot \mathbf{f}_j = 0$ for all $i \neq j$. We calculate that

$$\begin{aligned} \mathbf{f}_1 \cdot \mathbf{f}_2 &= 1 \cdot 1 + 1 \cdot 0 \\ &\quad + 1 \cdot 1 + (-1) \cdot 2 = 1 + 0 + 1 - 2 = 0 \end{aligned}$$

$$\mathbf{f}_1 \cdot \mathbf{f}_3 = 1 \cdot (-1) + 1 \cdot 0 + 1 \cdot 1 + (-1) \cdot 0 = -1 + 0 + 1 + 0 = 0$$

$$\mathbf{f}_1 \cdot \mathbf{f}_4 = 1 \cdot (-1) + 1 \cdot 3 + 1 \cdot (-1) + (-1) \cdot 1 = -1 + 3 - 1 - 1 = 0$$

$$\mathbf{f}_2 \cdot \mathbf{f}_3 = 1 \cdot (-1) + 0 \cdot 0 + 1 \cdot 1 + 2 \cdot 0 = -1 + 0 + 1 + 0 = 0$$

$$\mathbf{f}_2 \cdot \mathbf{f}_4 = 1 \cdot (-1) + 0 \cdot 3 + 1 \cdot (-1) + 2 \cdot 1 = -1 + 0 - 1 + 2 = 0$$

$$\mathbf{f}_3 \cdot \mathbf{f}_4 = (-1) \cdot (-1) + 0 \cdot 3 + 1 \cdot (-1) + 0 \cdot 1 = 1 + 0 - 1 + 0 = 0.$$

Since all these dot products are zero, the set is orthogonal.

To normalize these vectors, we need to calculate their lengths and divide each vector by its length. We find that

$$\|\mathbf{f}_1\| = \sqrt{1^2 + 1^2 + 1^2 + (-1)^2} = \sqrt{4} = 2$$
$$\|\mathbf{f}_2\| = \sqrt{1^2 + 0^2 + 1^2 + 2^2} = \sqrt{6}$$
$$\|\mathbf{f}_3\| = \sqrt{(-1)^2 + 0^2 + 1^2 + 0^2} = \sqrt{2}$$
$$\|\mathbf{f}_4\| = \sqrt{(-1)^2 + 3^2 + (-1)^2 + 1^2} = \sqrt{12} = 2\sqrt{3}.$$

Therefore the normalized vectors $\mathbf{q}_i = \mathbf{f}_i / \|\mathbf{f}_i\|$ are given by:

$$\mathbf{q}_1 = \frac{1}{2}\mathbf{f}_1 = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}$$

$$\mathbf{q}_2 = \frac{1}{\sqrt{6}}\mathbf{f}_2 = \begin{bmatrix} \frac{1}{\sqrt{6}} \\ 0 \\ \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \end{bmatrix}$$

$$\mathbf{q}_3 = \frac{1}{\sqrt{2}}\mathbf{f}_3 = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}$$

$$\mathbf{q}_4 = \frac{1}{2\sqrt{3}}\mathbf{f}_4 = \begin{bmatrix} -\frac{1}{2\sqrt{3}} \\ \frac{3}{2\sqrt{3}} \\ -\frac{1}{2\sqrt{3}} \\ \frac{1}{2\sqrt{3}} \end{bmatrix}.$$

Theorem: Pythagoras' theorem

If $\{\mathbf{f}_1, \mathbf{f}_2, \ldots, \mathbf{f}_k\}$ is an orthogonal set in $\mathbb{R}^n$, then

$$\|\mathbf{f}_1 + \mathbf{f}_2 + \cdots + \mathbf{f}_k\|^2 = \|\mathbf{f}_1\|^2 + \|\mathbf{f}_2\|^2 + \cdots + \|\mathbf{f}_k\|^2.$$

Theorem: Linear independence of orthogonal sets

Every orthogonal set in $\mathbb{R}^n$ is linearly independent.

Why is this true?

Let $\{\mathbf{f}_1, \mathbf{f}_2, \ldots, \mathbf{f}_k\}$ be an orthogonal set in $\mathbb{R}^n$. To show the set is linearly independent (recalling Linear Independence), we must show the only solution to the linear equation

$$t_1\mathbf{f}_1 + t_2\mathbf{f}_2 + \cdots + t_k\mathbf{f}_k = \mathbf{0}$$

occurs when $t_1 = \cdots = t_k = 0$.
If we take the dot product of both sides of the equation with $\mathbf{f}_1$, we find that

$$\mathbf{f}_1 \cdot (t_1\mathbf{f}_1 + t_2\mathbf{f}_2 + \cdots + t_k\mathbf{f}_k) = \mathbf{f}_1 \cdot \mathbf{0}$$

and therefore that

$$t_1(\mathbf{f}_1 \cdot \mathbf{f}_1) + t_2(\mathbf{f}_1 \cdot \mathbf{f}_2) + \cdots + t_k(\mathbf{f}_1 \cdot \mathbf{f}_k) = 0$$

Since the set of vectors is orthogonal, $\mathbf{f}_1 \cdot \mathbf{f}_j = 0$ for all $j \neq 1$, and $\mathbf{f}_1 \cdot \mathbf{f}_1 = \|\mathbf{f}_1\|^2$, so we conclude that

$$t_1\|\mathbf{f}_1\|^2 = 0.$$

Since $\mathbf{f}_1 \neq \mathbf{0}$ (by definition, an orthogonal set cannot contain the zero vector), $\|\mathbf{f}_1\|^2 > 0$, and so it must be true that $t_1 = 0$. But now repeating

this argument with 1 replaced by any index i, we conclude that $t_i = 0$. Therefore the set $\{\mathbf{f}_1, \dots, \mathbf{f}_k\}$ is linearly independent.

The expansion theorem

Theorem: Expansion theorem

Let $\{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_m\}$ be an orthonormal basis of a subspace U of $\mathbb{R}^n$. If $\mathbf{x}$ is any vector in U, we have

$$\mathbf{x} = (\mathbf{x} \cdot \mathbf{q}_1)\mathbf{q}_1 + (\mathbf{x} \cdot \mathbf{q}_2)\mathbf{q}_2 + \cdots + (\mathbf{x} \cdot \mathbf{q}_m)\mathbf{q}_m.$$

Why is this true?

Since $\{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_m\}$ is a basis of U, any $\mathbf{x} \in U$ can be written uniquely as

$$\mathbf{x} = c_1\mathbf{q}_1 + c_2\mathbf{q}_2 + \cdots + c_m\mathbf{q}_m$$

for some $c_1, \dots, c_m$. To find coefficient c_1, take the dot product of both sides with $\mathbf{q}_1$, obtaining that

$$\begin{aligned} \mathbf{x} \cdot \mathbf{q}_1 &= (c_1\mathbf{q}_1 + c_2\mathbf{q}_2 + \cdots + c_m\mathbf{q}_m) \cdot \mathbf{q}_1 \\ &= c_1(\mathbf{q}_1 \cdot \mathbf{q}_1) + c_2(\mathbf{q}_2 \cdot \mathbf{q}_1) + \cdots + c_m(\mathbf{q}_m \cdot \mathbf{q}_1). \end{aligned}$$

Since the basis is orthonormal, $\mathbf{q}_i \cdot \mathbf{q}_j = 0$ for $i \neq j$ and $\mathbf{q}_1 \cdot \mathbf{q}_1 = 1$, so the equation simplifies to

$$\mathbf{x} \cdot \mathbf{q}_1 = c_1.$$

A similar argument shows that

$$c_i = \mathbf{x} \cdot \mathbf{q}_i$$

for each i, concluding the proof.

Warning: Normalize before using the expansion formula

The formula in Expansion theorem requires an *orthonormal* basis: mutual orthogonality alone is not enough. If you are given only an orthogonal set $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$, first normalize each vector:

$$\mathbf{q}_i = \frac{\mathbf{f}_i}{\|\mathbf{f}_i\|}, \qquad i = 1, \dots, m.$$

The vectors $\mathbf{q}_i$ form an orthonormal basis of the same span. Apply the formula using these unit vectors, and remember that the vector being expanded must lie in their span.

Note: Coordinate vector in an ordered orthonormal basis

Let $B = (\mathbf{q}_1, \dots, \mathbf{q}_m)$ be an ordered orthonormal basis of U, and let $\mathbf{u} \in U$. Recall that the coordinate vector of $\mathbf{u}$ with respect to B lists its coefficients in this basis. By Expansion theorem,

$$C_B(\mathbf{u}) = \begin{bmatrix} c_1 \\ \vdots \\ c_m \end{bmatrix} = \begin{bmatrix} \mathbf{u} \cdot \mathbf{q}_1 \\ \vdots \\ \mathbf{u} \cdot \mathbf{q}_m \end{bmatrix}.$$

The order of the entries follows the chosen order of the basis vectors.

Activity: Expansion with two orthogonal vectors (U4-LO1)

Consider the vectors

$$\mathbf{f}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad \mathbf{f}_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \qquad \mathbf{u} = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$$

in $\mathbb{R}^3$, and let $U = \operatorname{span}\{\mathbf{f}_1, \mathbf{f}_2\}$.

1. Find the coefficients t_1, t_2 such that $\mathbf{u} = t_1\mathbf{f}_1 + t_2\mathbf{f}_2$ by the Anatomy of a linear map method used in Testing membership in a span (U2-LO4). Set up the augmented matrix, apply The Gaussian Elimination Algorithm, and use back substitution.

Activity: Expansion with two orthogonal vectors (U4-LO1) (continued)

Show the intermediate matrices and label each row operation using the shorthand from Row-operation shorthand.

2. Verify that $\{\mathbf{f}_1, \mathbf{f}_2\}$ is an orthogonal basis of U.
3. Normalize the basis vectors to obtain $\mathbf{q}_1, \mathbf{q}_2$, then compute the coordinates $c_i = \mathbf{u} \cdot \mathbf{q}_i$ and write the expansion of $\mathbf{u}$ in this orthonormal basis. Rewrite the expansion using the original vectors $\mathbf{f}_1, \mathbf{f}_2$ and compare with the first part. Check by direct addition that your linear combination equals $\mathbf{u}$.

1. Equating coordinates in $\mathbf{u} = t_1\mathbf{f}_1 + t_2\mathbf{f}_2$ gives

$$\begin{aligned} t_1 + t_2 &= 3, \\ t_1 - t_2 &= 1, \\ t_2 &= 1. \end{aligned}$$

The augmented matrix, with coefficient columns in the order t_1, t_2, is

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 1 & -1 & 1 \\ 0 & 1 & 1 \end{array}\right].$$

The first pivot is already a leading one. Elim-

inate below it, normalize the second pivot, and then eliminate below that pivot:

$$
\begin{aligned}
R_2 \to R_2 - R_1 \quad & \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & -2 & -2 \\ 0 & 1 & 1 \end{array}\right] \\
R_2 \to -\tfrac{1}{2}R_2 \quad & \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{array}\right] \\
R_3 \to R_3 - R_2 \quad & \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{array}\right].
\end{aligned}
$$

The last row is $0 = 0$, so the system is consistent. Both coefficient columns contain pivots, so there are no free variables. Back substitution gives $t_2 = 1$ and then $t_1 = 3 - t_2 = 2$. Thus

$$\mathbf{u} = 2\mathbf{f}_1 + \mathbf{f}_2,$$

which also shows that $\mathbf{u} \in U$.

2. Both basis vectors are nonzero, and

$$\mathbf{f}_1 \cdot \mathbf{f}_2 = 1 \cdot 1 + 1 \cdot (-1) + 0 \cdot 1 = 0.$$

Thus the set is orthogonal and therefore linearly independent. It spans U by definition, so it is an orthogonal basis of U.

3. The squared lengths are

$$\|\mathbf{f}_1\|^2 = 1^2 + 1^2 + 0^2 = 2,$$
$$\|\mathbf{f}_2\|^2 = 1^2 + (-1)^2 + 1^2 = 3.$$

Normalize first:

$$\mathbf{q}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad \mathbf{q}_2 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}.$$

The vectors remain orthogonal and now have length one. The coordinates of $\mathbf{u}$ with respect to this orthonormal basis are

$$c_1 = \mathbf{u} \cdot \mathbf{q}_1 = \frac{3+1}{\sqrt{2}} = 2\sqrt{2},$$
$$c_2 = \mathbf{u} \cdot \mathbf{q}_2 = \frac{3-1+1}{\sqrt{3}} = \sqrt{3}.$$

By Expansion theorem,

$$\begin{aligned}\mathbf{u} &= 2\sqrt{2}\,\mathbf{q}_1 + \sqrt{3}\,\mathbf{q}_2 \\ &= 2\sqrt{2}\left(\frac{\mathbf{f}_1}{\sqrt{2}}\right) + \sqrt{3}\left(\frac{\mathbf{f}_2}{\sqrt{3}}\right) = 2\mathbf{f}_1 + \mathbf{f}_2.\end{aligned}$$

Checking the resulting expansion gives

$$2\mathbf{f}_1 + \mathbf{f}_2 = 2\begin{bmatrix}1\\1\\0\end{bmatrix} + \begin{bmatrix}1\\-1\\1\end{bmatrix} = \begin{bmatrix}3\\1\\1\end{bmatrix} = \mathbf{u}.$$

The coefficients with respect to the original basis are $t_1 = 2$ and $t_2 = 1$, agreeing with Gaussian elimination in the first part. These differ from the coefficients $c_1 = 2\sqrt{2}$ and $c_2 = \sqrt{3}$ in the normalized basis.

Activity: Expansion in an orthonormal basis (U4-LO1)

Let $\mathbf{u} = (a, b, c, d)$ be a vector in $\mathbb{R}^4$. Normalize the orthogonal basis $\{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3, \mathbf{f}_4\}$ given in Checking orthogonality and normalizing vectors (U4-LO1) to obtain an orthonormal basis $\{\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3, \mathbf{q}_4\}$. Find the expansion of $\mathbf{u}$ in this orthonormal basis. Recall that

$$\mathbf{f}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix} \quad \mathbf{f}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 2 \end{bmatrix} \quad \mathbf{f}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad \mathbf{f}_4 = \begin{bmatrix} -1 \\ 3 \\ -1 \\ 1 \end{bmatrix}.$$

We first normalize the basis. In Checking orthogonality and normalizing vectors (U4-LO1) we already computed its lengths:

$$\|\mathbf{f}_1\| = 2, \qquad \|\mathbf{f}_2\| = \sqrt{6},$$
$$\|\mathbf{f}_3\| = \sqrt{2} \quad \text{and} \quad \|\mathbf{f}_4\| = 2\sqrt{3}.$$

Therefore,

$$\mathbf{q}_1 = \frac{\mathbf{f}_1}{2}, \qquad \mathbf{q}_2 = \frac{\mathbf{f}_2}{\sqrt{6}},$$
$$\mathbf{q}_3 = \frac{\mathbf{f}_3}{\sqrt{2}}, \qquad \mathbf{q}_4 = \frac{\mathbf{f}_4}{2\sqrt{3}}.$$

These vectors form an orthonormal basis of $\mathbb{R}^4$. The coefficients in the expansion of $\mathbf{u}$ are the dot products with these unit vectors:

$$\begin{aligned} c_1 = \mathbf{u} \cdot \mathbf{q}_1 &= \frac{a \cdot 1 + b \cdot 1 + c \cdot 1 + d \cdot (-1)}{2} \\ &= \frac{a + b + c - d}{2}, \end{aligned}$$

$$\begin{aligned} c_2 = \mathbf{u} \cdot \mathbf{q}_2 &= \frac{a \cdot 1 + b \cdot 0 + c \cdot 1 + d \cdot 2}{\sqrt{6}} \\ &= \frac{a + c + 2d}{\sqrt{6}}, \end{aligned}$$

$$\begin{aligned} c_3 = \mathbf{u} \cdot \mathbf{q}_3 &= \frac{a \cdot (-1) + b \cdot 0 + c \cdot 1 + d \cdot 0}{\sqrt{2}} \\ &= \frac{-a + c}{\sqrt{2}}, \end{aligned}$$

$$\begin{aligned} c_4 = \mathbf{u} \cdot \mathbf{q}_4 &= \frac{a \cdot (-1) + b \cdot 3 + c \cdot (-1) + d \cdot 1}{2\sqrt{3}} \\ &= \frac{-a + 3b - c + d}{2\sqrt{3}}. \end{aligned}$$

By Expansion theorem, the expansion in this orthonormal basis is

$$\mathbf{u} = \frac{a+b+c-d}{2}\mathbf{q}_1 + \frac{a+c+2d}{\sqrt{6}}\mathbf{q}_2 + \frac{-a+c}{\sqrt{2}}\mathbf{q}_3 + \frac{-a+3b-c+d}{2\sqrt{3}}\mathbf{q}_4.$$

Dot products and orthogonality now become tools for finding closest vectors.